

# NICK MCCULLOUGH

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## EDUCATION

### Iowa State University

Ames, IA

B.S. Software Engineering, 3.42 GPA

Expected Graduation: 12/2025

- Awards: Arthur A. Collins (Engineering Leadership) | John Hoyt Chaloud (Engineering) | 3x Engineering Dean's List
- Leadership: Founded tech-focused Discord community to help 125+ Iowa State students with coursework, internships, career growth. Directly assisted 10+ students in securing internships, indirectly assisted many more
- Coursework: Data Structures, Algorithm Analysis, Software Evolution, Software Architecture, Operating Systems

## TECHNICAL SKILLS

### Programming Languages/Tools

Python, JavaScript, Java, C, Ada, SQL / JIRA, git, Linux, MySQL

### Frameworks/Libraries

React.js, Node.js, Express.js, Spring Boot, TailwindCSS, npm

## PROFESSIONAL EXPERIENCE

### RTX - Collins Aerospace

Cedar Rapids, IA / Remote

Software Engineer Intern | Military Avionics Flight Management | Performance Software Team

May 2025 - Aug 2025

- Collaborated with principal software engineers to develop and maintain embedded Ada avionics performance software
- Completed software JIRA tickets, held official reviews, updated official software design documents for customer releases

Software Engineer Part-Time | Military Avionics Flight Management | Nav. & Guidance Software Team

Aug 2024 - May 2025

- Engineered scalable Python test automation for multiple military avionics programs, enabling single and multi-sensor GPS validation across aircraft systems, aligning testing SQL queries, providing robust test coverage options for 100+ engineers
- Debugged and fixed several existing embedded Ada software issues identified during ad-hoc testing by collaboratively analyzing root causes and potential solutions, enhancing safety-critical software of military aircraft worldwide
- Developed and fixed hundreds of Python avionics test procedures to identify bugs and ensure requirements are met

Software Engineer Co-Op | Military Avionics Flight Management | Nav. & Guidance Software Team

Jan 2024 - Aug 2024

- Partnered with principal software engineers in a team to develop and maintain Ada navigation and guidance software
- Increased CH-47F program bidding accuracy 36% by creating a Python-based HTML parser using efficient algorithms to find logical Ada code blocks, prioritizing complexity over line count, ensuring accurate time needed for failed test reviews
- Designed in-air guidance pattern algorithms to optimize flight paths for military aircraft mission safety and efficiency
- Implemented satellite WAAS compatibility updates for navigation software, enhancing GPS solution accuracy & integrity
- Programs: E-3 Sentry, UH-60 Blackhawk, MH-6 Little Bird, E-2D Hawkeye, KC-135 Tanker, HC-144 Sentry, CH-47F Chinook

### John Deere

Ames, IA

Software Engineer Intern | Large Square Balers Product Engineering | GPS Sensor Team

Jul 2022 - Apr 2023

- Collaborated with senior engineers to develop and optimize L340/L341 tractor baler software using C in Agile sprints
- Optimized GPS algorithms, boosting bale location accuracy by 15%, delivering significant time/cost savings for farmers

## SOFTWARE ENGINEERING PROJECTS

**GridAI** | [Website](#) | Senior Design Research Project | React.js, TypeScript, Deck.GL, Maps, Docker

2025

- Led a software team of six, overseeing requirements and weekly deliverables, technical direction and collaboration
- Independently developed a responsive front-end Map Box geospatial interface for large datasets using Deck.GL, React MapGL and Dexie.js, integrating React hooks with RESTful APIs to visualize real-time and historical grid data for customers
- Optimized real-time power grid data using DexieJS with efficient queries, web-based data rendering, caching and layering

**Autonomous Search & Destroy Robot** | [GitHub](#) | Embedded C, PuTTY, Code Composer Studio, TM4C, iRobot Create 2

2025

- Developed a fully autonomous robot in C, simulating a round of Search and Destroy inspired by the game VALORANT
- Robot iteratively moves and scans surroundings using servo-mounted PING & ADC(IR) sensors, UART for communication, obstacle and map boundary avoidance, teammate and enemy object width detection, locates planting area to win game

**Movie Magnet - Android App** | [GitHub](#) | [Video](#) | [Awards](#) | Java, Spring Boot, Maven, MySQL

2023

- Developed a movie-lovers mobile app with theater finder, age groups, reviews, real-time chatting and admin features
- Played a crucial role in driving the project from idea to completion, balancing back-end development and team leadership

**Music Merchandise Full-Stack Website** | [GitHub](#) | [Video](#) | [Awards](#) | React.js, Tailwind CSS, Node, Express, MySQL

2023

- Developed a music merchandise storefront with React, featuring advanced filtering, ratings, and a seamless cart system
- Created a dynamic, responsive front-end with integrating RESTful APIs with MySQL for efficient data retrieval